



Péter Juhász, PhD

Quantitative Researcher

CONTACT

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LinkedIn

EDUCATION

2022 – 2025 PhD in Mathematics
Aarhus University, Denmark

2019 – 2021 MBA in Finance
Budapest University of Technology and Economics, Hungary

2013 – 2016 MSc in Physics
Budapest University of Technology and Economics, Hungary

2013 – 2016 BSc in Physics
Budapest University of Technology and Economics, Hungary

SKILLS

Modeling: Stochastic Models, Probability, Data Analysis

Programming: Python, C++, SQL

Tools: Git

LANGUAGES

English
German
Hungarian

CERTIFICATIONS

2021 Software Architect ISAQB

2021 Software Designer Bosch

PROFILE

I am a PhD in Mathematics with experience in stochastic modeling, probabilistic analysis, and large-scale simulations with a background in Python/C++ model development. Previously, I was a machine learning researcher at Bosch, where I developed and validated predictive models focusing on uncertainty estimation and data-driven decision-making. I am interested in applying quantitative methods to forecasting and risk modeling in energy and financial markets.

PROFESSIONAL EXPERIENCE

PhD Researcher in Mathematics **2022 – 2025**
Aarhus University, Denmark

- Developed and analyzed stochastic models and statistical tests for high-dimensional random systems.
- Designed and implemented high-performance Monte Carlo algorithms in Python and C++ for hypothesis testing and validation of models against datasets.
- Taught advanced probability, data analysis, and ODEs to mathematics and data-science students.

Machine Learning Engineer **2018 – 2022**
Bosch Group, Hungary

- Patented a Gaussian regression based trajectory-prediction model, improving predictive accuracy by 3× and uncertainty calibration by 10× over baseline.
- Built prototypes in Python for forecasting and uncertainty quantification, and translated them into production environments.
- Conducted large-scale data analysis and validation experiments to evaluate model performance.

Software Engineer **2016 – 2018**
Ericsson Telecommunications, Hungary

- Analyzed response times of distributed systems to optimize load-balancing strategies.
- Implemented performance-critical algorithms for Smart Services Routers.

AWARDS, GRANTS

2022 PhD Fellowship

Danish Data Science Academy
1 890 000 DKK

2021 Excellence Award

Hungarian National Bank
350 000 HUF

2025 Individual Travel Grant

European Mathematical Society
500 EUR

2024 Academic Mobility Grant

Aarhus University
20 000 DKK

SELECTED PUBLICATIONS, PATENTS

- C. Hirsch, B. Jahnel, and P. Juhász. **2025**
Functional limit theorems for edge counts in dynamic random connection hypergraphs
- P. Juhász. **2021**
Method and device for predicting the trajectory of a traffic participant, and sensor system
- P. Juhász. **2021**
Information propagation in stochastic networks
- P. Juhász, J. Stéger, D. Kondor, and G. Vattay. **2018**
A Bayesian approach to identify Bitcoin users